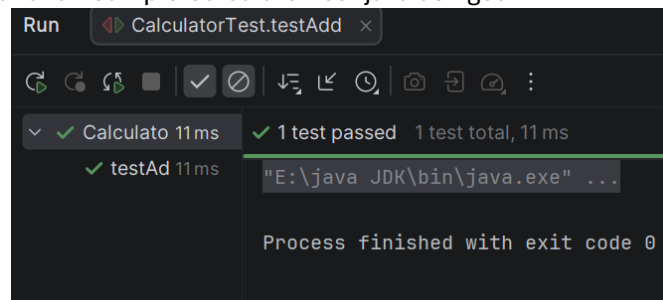


Test Driven Development using JUnit5

Exercise 1: Setting Up Junit

Added Dependencies and ran Sample CalculatorTest.java using Junit



Junit Dependency in pom.xml

```
<dependency>
  <groupId>org.junit.jupiter</groupId>
  <artifactId>junit-jupiter-api</artifactId>
  <version>5.11.4</version>
  <scope>test</scope>
</dependency>
```

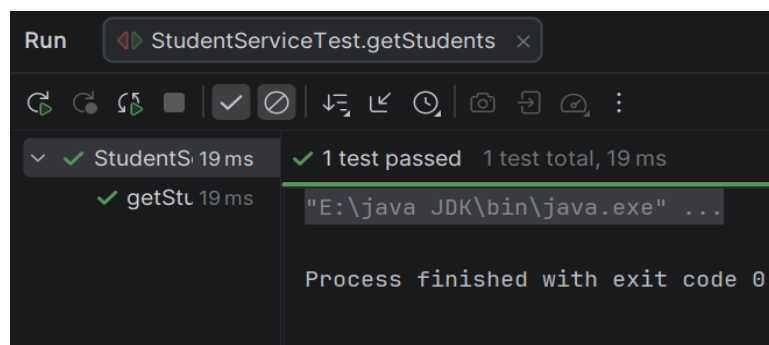
Exercise 3: Assertions in JUnit

Created Student class which accepts Student details id and name

Created StudentService Class to store the list of students

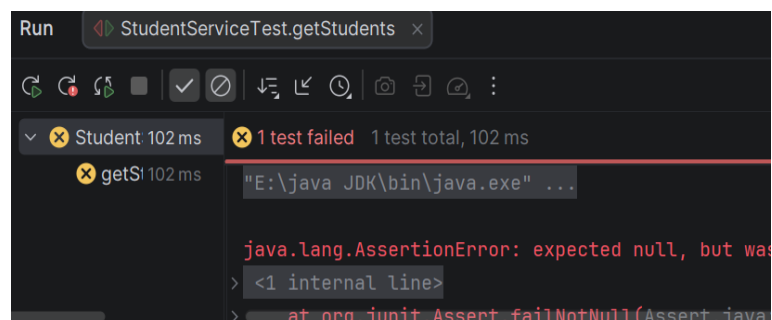
All assertions are Executed and checked

All assertions passed



So checked with different data for Test Failing Case

Example Student student =new Student(1,"Sonu");
assertNull(student);



Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in Junit

Junit Dependency in pom.xml

```
<dependency>
  <groupId>org.junit.jupiter</groupId>
  <artifactId>junit-jupiter-api</artifactId>
  <version>5.11.4</version>
  <scope>test</scope>
</dependency>
```

Created CartService Class with methods

- addItem()
- getItemCount()
- clearCart()

Created Test Class for CartService

Used @BeforeEach and @AfterEach Annotations with setup() and teardown() methods

3 Test Methods

- addItem()
- getItemCount()
- clearCart()

Output:

